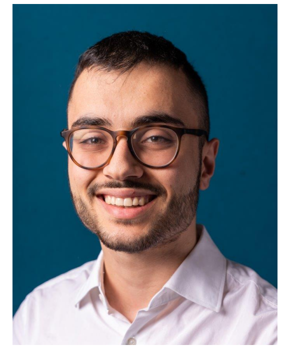
Javier Páez Franco

- business.antler474@passinbox.com
- linkedin.com/in/javier-paez-franco
- github.com/umenzi
- jpaefra.com



Research Interests

- **Robotics**: Manipulation, Locomotion, Safe Autonomous Control, Assistive Robotics, Human-Robot Interaction, Perception
- **Machine Learning**: Reinforcement, Imitation, & Deep Learning, World Models, Computer Vision, Continual Learning

Education

Norwegian University of Science and Technology — Jun. 2026 – Present
PhD in Autonomous Robotics — *Trondheim, Norway*

- **Goal**: Researching mechanisms for a shared autonomy architecture that generalises across embodiments, by making use of foundation models, predictive control, and robotic world models. Supervised by Prof. Dr. Konstantinos Alexis.

Imperial College London — Sep. 2024 – Sep. 2025
M.Sc. Computing (Artificial Intelligence and Machine Learning) — *London, UK*

- **Grade**: Distinction
- **Focus**: Robotics, Reinforcement & Imitation Learning, Deep Learning, Computer Vision, Mathematics, Optimisation

Delft University of Technology — Sep. 2021 – July 2024
B.Sc. Computer Science and Engineering — *Delft, Netherlands*

- **Grade**: *Cum Laude* with Honors
- **Honours Programme**: 2022-2024; 20 additional ECTS in Robotics Research
- **Minor in Robotics**: Courses in electronic circuits, Physics & dynamics, design, planning & decision-making

CERN IdeaSquare Summer School — May 2023 – July 2023
Summer Programme — *Delft, Netherlands – Geneva, Switzerland*

- Selected as one of the 20 students from the leading universities in the Netherlands.
- Collaborated with an interdisciplinary team to create **UltraGPU**, which is 50% more energy-efficient and has a 5000% longer lifespan than typical GPUs, significantly accelerating neural network training.

Research Experience

Non-Stationary Reinforcement Learning: Gaussian Proximal Policy Optimisation — Feb. 2025 – Sep. 2025

- Distinction-awarded Master Thesis supervised by Dr. Bahadir Kocer and Dr. Melih Kandemir.
- Developed a novel Reinforcement Learning algorithm that employs the principled uncertainty quantification of a Gaussian Process critic to allow robust adaptation in non-stationarity, preserving plasticity.
- Submitted for publication.

Addressing Diversity in Reinforcement Learning from Human Feedback (RLHF) — May 2023 – July 2024

- 9.0/10.0 Bachelor Thesis supervised by Dr. L. Cavalcante Siebert, available on the TU Delft Repository.
- Showed that RLHF performance rapidly degrades with even minimal conflicting feedback, and current query selection strategies are ineffective in handling feedback diversity.

Bio-inspired Navigation of Multi-Agent Systems in Extreme Environments — Nov. 2023 – July 2024

- Benchmarked bio-inspired algorithms for multi-agent path planning, optimising for reachability, planning time, and path length under computational constraints. Supervised by Dr. Raj Thilak Rajan and Prof. Martijn Wisse.

How Educational Techniques Foster Creativeness and Innovative Thinking — Jun. 2023 – Oct. 2023

- Written a paper in collaboration with CERN, researching how creativity is impacted within educational teams.
- Published in the CERN IdeaSquare Journal of Experimental Innovation (CIJ).

Distributed Algorithm to Maximize the Lifetime of a Swarm of Rovers — Dec. 2022 – Oct. 2023

- Developed a distributed algorithm to maximise the lifespan of a rover swarm for a 14-day lunar mission. Collaborated with Lunar Zebro, a team pioneering cutting-edge lunar rover technology.

Work experience

SHV Energy – Autonomous robot
Sep. 2023 – Feb. 2024

Robotics Engineer Intern — *Delft, Netherlands*

- Led an interdisciplinary team of six students to build a robot to autonomously load LPG cylinders onto a truck.
- Reduced the manual handling risk by loading and unloading over 1200 cylinders per hour.
- Leveraged Computer Vision and LiDAR sensors to accurately detect cylinders and ensure safety.
- Developed the robotic system using Python, ROS2, Computer Vision, and Rust, integrating Gazebo for simulation.

Your Next Agency – Project Monitoring Dashboard SaaS
May 2023 – July 2023

Software Engineer Intern — *Amsterdam, Netherlands*

- Collaborated within a five-person team to develop a management dashboard SaaS application.
- Automated 70% of project creation and user management, becoming the company's primary portal.
- Implemented using PHP, Laravel, TypeScript, Node.js, AWS, and PostgreSQL.

Panacea Cooperative Research – Skeleton ID
July 2022 – Sep. 2022

Software Engineer Intern — *Ponferrada, Spain*

- Actively developed Skeleton-ID, a forensic identification software powered by Deep Learning. Collaborated closely with anthropologists and researchers, providing essential support to drive successful project outcomes.
- Demonstrated first-hand knowledge of research methodology through active involvement in multiple research initiatives.

Selected Projects

Real-Time ML Inference System | *Python, Scikit-Learn, NumPy, Docker, Kubernetes* — Dec. 2024 – March 2025

- Developed a Machine Learning system to predict Acute Kidney Injury (AKI) in real time with minimal latency.
- Achieved an F3 score of 97%, significantly outperforming the industry-standard algorithms, which achieve only ∼70%.
- Deployed using Kubernetes and Docker, adhering to the HL7 data standards.

Computer Vision Model: Masked Auto-Encoders | *Python, PyTorch, Scikit-Learn* — Feb. 2025

- Collaborated with a peer to train a Masked Auto-Encoder (MAE) model on the PneumoniaMNIST imaging dataset.
- Analysed the learned embeddings using t-SNE to visualise and interpret the model's feature representations. The model successfully extracted discriminative features, clustering together data of the same category in the embedding space.

Imitation Learning for Robotic Navigation | *Python, PyTorch, NumPy* — Feb. 2025

- Developed an autonomous agent that learned to navigate dynamic terrains through Behavioural Cloning, achieving 60% success rate in unseen environments.
- Designed a multi-heuristic system to intelligently request new demonstrations, optimising the learning process.

Generative AI Model: Diffusion | *Python, PyTorch, Scikit-Learn, Hugging Face* — Jan. 2025 – Feb. 2025

- Developed a Diffusion Model (DDPM) in PyTorch from scratch to generate realistic hot dog images.
- Used sinusoidal position embeddings, an UNet architecture, and a pretrained VAE encoder (from HuggingFace).

Ray Tracing Engine | *C++, OpenGL* — Jan. 2023 – Feb. 2023

- Developed a ray tracing engine within a group of three students.
- Implemented shading, scene lighting, texture support, motion blur, and environment maps.

Certifications

- **ROS1x: Hello (Real) World with ROS – Robot Operating System:**

Extracurricular Activities

Krashna Musika
Oct. 2021 – July 2024

First violinist — *Delft, Netherlands*

- First violin in Krashna Musika, the symphonic orchestra and choir of TU Delft. With over 200 members, Krashna Musika is one of the largest musical student associations in the Netherlands.
- Played concerts internationally, such as in The Netherlands and France.

Skills

Programming languages: Python, C, C++, Java, Rust, SQL, Scala, JavaScript, PHP, Kotlin, Bash
Frameworks & Tools: PyTorch, Scikit-learn, ROS, Gazebo, NumPy, OpenCV, Docker, Kubernetes, Git, Unix
Languages: English (IELTS C1), Spanish (Native)

CV last updated: June 2026